COURSE STRUCTURE FOR B.TECH. Mechanical Engineering THIRD YEAR SEM – V (w. e. f. 2024-25)

SEMESTER-V (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-V											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total Marks
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	
1	HSC	24HS301T	Engineering Economics	3	0	0	3	3	25	25	50	-	-	100
2	OE	MOOC	NPTEL / SWAYAM / MOOC Course	3	0	0	3	3	25	25	50	-	-	100
3	PE	24ME331T	Introduction to Composite Materials	3	0	0	3	3	25	25	50	-	-	100
		24ME332T	Renewable and Sustainable Energy Technologies						25	25	50	-	-	100
		24ME333T	Fluid Machinery						25	25	50	-	-	100
		24ME334T	Laser and Electron Beam Material Processing						25	25	50	-	-	100
4	PCC	24ME301T	Heat Transfer	3	0	0	3	3	25	25	50	-	-	100
5	PCC	24ME301P	Heat Transfer Laboratory	0	0	2	1	2	-	-	-	25	25	50
6	PCC	24ME302T	Dynamics of Machine	3	0	0	3	3	25	25	50	-	-	100
7	PCC	24ME302P	Dynamics of Machine Laboratory	0	0	2	1	2	-	-	-	25	25	50
8	PCC	24ME303T	Manufacturing Processes - I	3	0	0	3	3	25	25	50	-	-	100
9	PCC	24ME303P	Manufacturing Processes - I Laboratory	0	0	2	1	2	-	-	-	25	25	50
			Total	**18**	**0**	**6**	**21**	**24**						

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME331T					Introduction to Composite Materials					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of composites, types, and properties.
2. To learn the role and selection of reinforcement and matrix
3. To study the various fabrication methods of composites.
4. To study advanced applications of composites.

UNIT I: INTRODUCTION 10 Hrs.

Classifications of Engineering Materials, Concept of composite materials, Matrix materials, Functions of a Matrix, Desired Properties of a Matrix, Types of matrix, Types of Reinforcements/Fibers: Role and Selection or reinforcement materials, Types of fibres, Mechanical properties of fibres. Material properties that can be improved by forming a composite material and its engineering potential.

UNIT II: COMPOSITE CLASSIFICATION 10 Hrs.

Classification based on Matrix Material: Organic Matrix composites, Polymer matrix composites (PMC), Metal matrix composites (MMC), Ceramic matrix composites (CMC); Classification based on reinforcements: Fiber Reinforced Composites, Fiber Reinforced Polymer (FRP) Composites, Laminar Composites, Particulate Composites, Comparison with Metals, Advantages & limitations of Composites, Isotropic, Anisotropic, symmetric, Unsymmetric laminates, Introduction to micromechanics of composites and failure criterion.

UNIT III: MANUFACTURING PROCESSES OF COMPOSITES 10 Hrs

Manufacturing Processes: Selection and Considerations, Hand Lay-up, Prepreg Lay-up, Bag Molding, Autoclave Processing, Compression Molding, Resin Transfer Molding, Vacuum Assisted Resin Transfer Molding, Pultrusion, Filament Winding, Carbon matrix Composites or Carbon-Carbon Composites, Nanocomposites.

UNIT IV: ADVANCED APPLICATION OF COMPOSITES 10 Hrs.

Aircraft: Composite Components in Aircraft, Specific Aspects of Structural Resistance, Large Carriers, Regional Jets, Light Aircraft, Fighter Aircraft, Turbine Blades in Composites, Helicopters, Composite Materials and the Manufacturing of Automobiles, Composites in Naval Construction, Sports and Recreation.

TOTAL HOURS: 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall the fundamental of metals and composites.
CO2 : Explain the selection and role fibres and matrix and its properties.
CO3 : Demonstrate different advanced application of composites.
CO4 : Apply the various fabrication techniques to manufacture composites.
CO5 : Classify the different types of composites.
CO6 : Decide specific type of composite applicable to different applications.

TEXT/REFERENCE BOOKS

1. Krishna K Chawla, "Composite Materials Science and Engineering", Springer.
2. B. D. Agrawal, "Analysis and Performance of Composites" Wiley.
3. Gay, Daniel, "Composite materials: design and applications" CRC press.
4. "Materials characterization", Vol. 10, ASM hand book

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME332T					Renewable and Sustainable Energy Technologies					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of basics of different sustainable type energy sources
2. To identify the sustainable energy technologies applicable in mechanical industries

UNIT I: SOLAR ENERGY **12 Hrs.**

Principles of solar radiation - physics of the sun, the solar constant, extra-terrestrial and terrestrial solar radiation, solar radiation on titled surface, instruments for measuring solar radiation and sun shine, solar radiation data, **Solar energy collection** - Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors, **Solar energy storage and applications** - Different methods, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating/cooling technique, solar distillation and drying, **Photovoltaic energy conversion** - Photovoltaic, p-n junction, solar cells, PV systems, Stand-alone, Grid connected solar power

UNIT II: BIOMASS ENERGY **10 Hrs.**

Biogas - Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, gas yield, combustion characteristics of bio-gas, utilization for cooking, **Biomass gasification** - Biomass conversion technologies, Constructional details of gasifier, **Biofuels** - Introduction and perspective of biofuels, biofuel production and applications, energy plantation, environmental impact of biofuel, Biofuel operated I.C. Engine operation and economic aspects, Biomass pallets and its applications

UNIT III: WIND ENERGY **10Hrs.**

Principle of wind energy conversion, Basic components of wind energy conversion systems - Lift and Drag– Effect of density, frequency variances, angle of attack, and wind speed - design considerations of horizontal and vertical axis wind machines - analysis of aerodynamic forces acting on wind turbine blades and estimation of power output - wind data and site selection considerations, speed control devices for wind turbine

UNIT IV: OTHER RENEWABLE ENERGY **10Hrs**

Energy from Ocean – Basic cycles of Ocean Thermal Energy Conversion, basic principle of tidal power, wave energy conversion devices, **Fuel Cells** - Introduction, Design principle and operation of fuel cell, Types of fuel cells, conversion efficiency of fuel cell, application of fuel cells, **Geothermal Energy** - nature of geothermal energy, geothermal sources and resources like hydrothermal, geo-pressured hot dry rock, magma. advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India, **Hydrogen Energy** - Introduction, Hydrogen Production methods, Hydrogen storage, hydrogen transportation, utilization of hydrogen gas, hydrogen as alternative fuel for vehicles

TOTAL HOURS: 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Understand** the principle of solar radiation and its availability at various locations
CO2 : **Evaluate** the ways to use solar energy for various applications like heating, cooling, water distillation and electricity and evaluate their performance.
CO3 : **Interpret** and **classify** the ways to harness the energy from Biomass.
CO4 : **Understand** and **analyze** the Wind and wave energy conversion.
CO5 : **Summarize** the ways to harness energy from other renewable energy sources like geothermal, ocean, Fuel cell, hydrogen etc.
CO6 : **Recognize** the need and ability to **engage** in lifelong learning for further developments in this field

TEXT/REFERENCE BOOKS

1. Sukhatme, S.P. and Nayak, J.K., **"Solar Energy",** McGraw-Hill Education.
2. Duffie, J.A., Beckman, W.A. and Worek, W.M., **"Solar Engineering of Thermal Processes (Vol. 3)",** New York: Wiley.
3. Zobaa, A.F. and Bansal, R.C., **"Handbook of Renewable Energy Technology"** World Scientific.
4. Kothari, D.P., Singal, K.C. and Ranjan, R., **"Renewable Energy Sources and Emerging Technologies",** PHI Learning Pvt. Ltd.
5. Desai, A.V., **"Nonconventional Energy",** New Age International.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME333T					Fluid Machinery					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand basics of the energy conversion in fluid machines
2. To understand working and performance of various machines.
3. How to compare and chose fluid machines for various operations.

UNIT I: FUNDAMENTALS OF FLUID MACHINES **8Hrs.**

Introduction: Definition and classification of turbo machines; Principles of operation; Energy transfer in turbo machines. Momentum Principles: Action of jets on stationary and moving flat plates and curved vanes; Angular momentum principle; Torque and efficiency of roto-dynamic machines

UNIT II: TURBINES **11Hrs.**

Classification; Impulse; Reaction; Radial, Axial, mixed and tangential flow turbines; Pelton, Francis turbines; Runner profiles; Velocity triangles; Head and efficiency; Draft tube theory; Similarity laws; Concept of specific speed and unit quantities; Selection of Turbines; Operational characteristics. Introduction to Turbines used in Renewable energy system such as Wind and Tidal Turbines

UNIT III: HYDRAULIC PUMPS **11Hrs.**

Working principles and classification of pumps; Centrifugal Pumps:; working principle of roto-dynamics pumps; Manometric head, Losses and efficiencies; velocity vector diagrams and work done; Priming; Performance and characteristic curves; NPSH and Cavitation; Similarity relations and specific speed; design considerations; multi-stage pumps. Reciprocating Pumps: Construction and operational details; work and power input; volumetric efficiency and slip; separation; air vessels and their utility; maximum speed of the rotating crank; characteristic curves, regenerative pumps and its application.

UNIT IV: AIR COMPRESSOR, FAN AND BLOWER **10Hrs.**

Air Compressor: Working principles and classification; Reciprocating air compressor: compression process, work of compression, single and multi-stage compression, volumetric efficiency, air motors. Rotary compressor: centrifugal and axial flow compressor, positive displacement compressor, velocity diagram, Analysis, Design and construction features. Compressor characteristics, surging and choking. Fans & Blowers: Working principles; types; velocity diagrams; stage parameters, design parameters, losses, screw compressor and its application. Integration of CFD: modules on CFD techniques for analysing the fluid flow within machinery.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 : Understand Fundamentals - Grasp energy conversion, classification, and operation of fluid machinery, including momentum principles and energy transfer in turbo machines
- CO2 : Analyze Turbines - Examine hydraulic turbines (Pelton, Francis, etc.) using velocity triangles, efficiency, and draft tube theory to understand design and operation.
- CO3 : Evaluate Pumps - Assess centrifugal and reciprocating pumps' principles, efficiencies, and performance, focusing on NPSH, cavitation, and air vessels.
- CO4 : Assess Compressors and Blowers - Analyze air compressors, fans, and blowers' classification, principles, and design, including efficiency and performance characteristics.
- CO5 : Synthesize for Selection - Use knowledge to evaluate and select suitable fluid machines for different operations, considering performance and operational characteristics
- CO6 : Create Optimized Designs - Develop optimized designs for turbines and pumps by incorporating multi-stage pumping, specific speed, and unit quantities to enhance efficiency.

TEXT/REFERENCE BOOKS

1. Shepherd, D. G., "**Principles of Turbo machinery**", Collier Macmillan.
2. S. L. Dixon, "**Fluid Mechanics, Thermodynamics of Turbomachinery**", Pergamon Press Ltd.
3. R. K. Turton, "**Principles of Turbomachinery**", Springer.
4. Wright, T., "**Fluid Machinery: Performance, Analysis and Design**", CRC Press.
5. S.M. Yahya, "**Turbines, Compressor and Fan**", McGraw-Hill, 2010.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME334T					Laser & electron beam Material Processing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	--	0	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To Introduce to practical engineering applications of Laser and electron(L & E) beam for material processing .
2. To **understand the** need and study the characteristics of L & E beam.
3. To study and appreciate different manufacturing processes processed by L & E beam.
4. To develop mathematical models for the L & E beam manufacturing processes and **Solve** them.

UNIT 1 — 10Hrs.

Main industrial laser and their output beam characteristics; laser beam delivery systems. Laser beam interaction with metal, semiconductor and insulator, Ultra-short laser pulse interaction, heat flow theory and metallurgical considerations. Laser Material Processing Applications: Laser. Fundamental Properties of Electron Beam; Generation and focussing of electron beam in Vacuum and non-vaccum Fundamental Properties of Electron Beam. Electron beam material processing application. Application of L & E beam to various manufacturing processes welding, cutting, drilling, etc. Process characteristics.

UNIT 2 — 10Hrs.

L & E beam welding: Process mechanisms like keyhole and plasma effect, operating characteristics and process variation. L & E surface modifications, Laser rapid manufacturing. Laser metal forming. Laser peening: Laser Shock Processing. Heat transfer and fluid flow, microstructure selection during the solidification processes, Residual, Solidification governing equations and microstructure selection. Phase-change. Nucleation and Growth kinetics, Rapid solidification

UNIT 3 — 12Hrs.

Theoretical and modelling of L & E beam material processing: Welding, Cutting, Drilling, surface modification, etc. On-line Process monitoring & control: sensors for process modelling and processing of sensor outputs for various laser and process parameters, and workpiece characteristics. L & E beam Safety. Mathematical modelling of different manufacturing processes with L & E beams, considerations in transport. N procedures to solve coupled mass, momentum, energy, and solute equation.

UNIT 4 — 12Hrs.

Additive manufacturing technologies/Rapid prototyping using L & E beam: Liquid-Based System, Powder based systems and solid based systems. Powder bed fusion, wire arc additive manufacturing, Direct material deposition using both beams: comparison of both the technologies. High energy density beams for 3D printing for metals, semiconductors, polymers, etc. Medical and nano technology applications for L & E beam material processing. Case-studies: Thermal simulations and analysis for manufacturing processes I & E beam material processing. Verification, validation and benchmarking. Inverse heat transfer problems. Microstructure modelling

Max. 44 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – To **understand and developing insight** into the physical phenomenon of laser-matter and electron beam interaction
CO2 – To **Study** Laser and electron beam technologies for material processing at micro macro scales.
CO3 – **Choose, Understand, implement** appropriate numerical solidification/melting model to solve problems.
CO4 – **Development** of first order mathematical descriptions for select laser & electron based manufacturing processes.
CO5 - **Understand the** capabilities and limitations of various processes processed by Laser and electron beam.
CO6 – **Choose** respective process parameters for appropriate process: Justify the process-parameter selection, design for manufacturing and quality control

TEXT/REFERENCE BOOKS

1. Kannatey-Asibu Jr, Elijah. *Principles of Laser Materials Processing: Developments and Applications*. John Wiley & Sons, 2023.
2. Steen, William M., and Jyotirmoy Mazumder. Laser material processing. springer science, 2010.
3. Gong, Shuili, et al. *Electron Beam Wire Deposition Technology and Its Application*. Springer, 2022.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME301T					Heat Transfer					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To develop the fundamental principles and laws of heat transfer.
2. To formulate the models necessary to study, analyse and design heat transfer systems through the application of these principles.
3. To develop the problem-solving skills essential to good engineering practice of heat transfer in real-world applications.

UNIT 1 CONDUCTION — 12 Hrs.

Introduction: Modes, mechanisms and basic concepts of heat transfer; Fourier's law, thermal conductivity, thermal diffusivity. **Heat Conduction:** Generalized (3D) heat conduction equation in the Cartesian, cylindrical and spherical coordinates; One dimensional steady state solutions (with and without heat generation); temperature profile and heat transfer equations; Boundary conditions; wall, cylinder, sphere and composites; electrical analogy; overall heat transfer coefficient; Variable thermal conductivity; Critical radius of insulation. **Extended surfaces:** Fin performance parameters. **Transient heat conduction:** Lumped system analysis; approximate analytical and graphical solutions for plane walls and semi-infinite solids.

UNIT 2 CONVECTION — 11 Hrs.

Heat Convection: Classification, physical mechanism and dimensional analysis applied to forced and free convection; local and average heat transfer coefficients and dimensionless numbers; Thermal and hydrodynamic boundary layers; Differential convection equations and solutions for flat plate; Analogies between momentum and heat transfer; Laminar and turbulent heat flow correlations for external flow. **Internal forced convection:** Concept of average velocity and temperature; entrance region; General thermal analysis for laminar flow in a tube; heat transfer correlations for laminar and turbulent flow. **Free convection:** Equation of motion and the Grashof number; natural convection over surfaces and inside enclosures; combined natural and forced convection. **Boiling and Condensation:** Boiling regimes and the boiling curve; Film and drop wise condensation; correlations. flow boiling

UNIT 3 RADIATION — 09 Hrs.

Thermal Radiation: Concept of radiation, absorptivity, reflectivity & transmissivity, blackbody, grey surfaces, emissive power & emissivity. Laws of radiation – Planck, Stefan – Boltzman, Wein's displacement, Kirchoff. Intensity of radiation & solid angle, Lambert's cosine law, shape factor. Radiation heat exchange between black bodies, heat exchange between diffuse-grey bodies-radiation shield, heat exchange between enclosed grey surfaces, electrical analogy to simple problems

UNIT 4 HEAT EXCHANGERS — 10 Hrs.

Heat Exchangers: Classification, heat exchange performance analysis, LMTD for parallel & counter flow heat exchanger, overall heat transfer coefficient, fouling, correction factor for multi-pass arrangement, effectiveness and number of transfer unit for parallel and counter flow heat exchanger, cross flow, TEMA standards.

Max. 42 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1: Comprehend modes of heat transfer and apply principles of heat transfer to solve engineering problems.
CO2: Analyze and Solve problems involving steady and unsteady heat conduction through solids.
CO3: Evaluate convection heat transfer problems for a variety of flow conditions using appropriate convection correlations.
CO4: Analyze heat exchanger performance using LMTD and NTU methods.
CO5: Evaluate radiative heat exchange between two or more surfaces of different geometries.
CO6: Understand the basic principles of heat transfer in boiling and condensation.

TEXT/REFERENCE BOOKS

1. Yunus A. Cengel and Afshin J. Ghajar, **"Heat and Mass Transfer: Fundamentals and Applications",** McGraw Hill Education. Fifth Edition, 2017.
2. F. P. Incropera and D. P. DeWitt, **"Introduction to Heat Transfer",** Wiley, , New York, 5th edition, 2006.
3. J. P. Holman, **"Heat Transfer",** Tata McGraw Hill Education, 10th Edition, 2017.
4. F. White, **"Heat and Mass Transfer",** Pearson Education (US), 1988.
5. S. P. Sukhatme**, "A Textbook of Heat Transfer",** Universities Press, 4th Edition, 2005.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME301P					Heat Transfer Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To analyse, differentiate and evaluate different modes of heat transfer through various mediums.
2. To evaluate steady and transient state properties of heat transfer mediums.
3. To calculate and compare the performance of heat exchangers.

LIST OF EXPERIMENTS

1. To Identify the Thermal Conductivity of Metal Rod
2. To Identify the Thermal Conductivity of Insulating Powder
3. To Identify the Convective Heat Transfer of Air In Natural Convection Mode
4. To Identify the Convective Heat Transfer of Air In Force Convection Mode
5. To Identify the Emissivity of A Test Material
6. To Verify the Stefan Boltzmann Constant
7. To Evaluate the Performance of Tube-In-Tube Heat Exchanger Under Parallel And Counter Flow Conditions
8. To Evaluate the Performance of Shell And Tube Heat Exchanger Under Various Flow Conditions
9. To Evaluate the Performance of Plate Heat Exchanger Under Various Flow Conditions

COURSE OUTCOMES

On completion of the course, student will be able to

CO1: Understand the engineering examples of different modes of heat transfer.
CO2: Evaluate and verify heat transfer modelling through experimentation.
CO3: Understand and analyse emissivity of a given materials.
CO4: Understand and analyse thermal conductivity of a given insulating powder.
CO5: Examine the performance of heat exchangers.
CO6: Compile and interpret the experimental data at steady state condition

TEXT/REFERENCE BOOKS

1. Yunus A. Cengel and Afshin J. Ghajar, **"Heat and Mass Transfer: Fundamentals and Applications,"** McGraw Hill Education. Fifth edition, 2017.
2. F. P. Incropera and D. P. DeWitt, **"Introduction to Heat Transfer",** Wiley, , New York, 5th edition, 2006.
3. J. P. Holman, **"Heat Transfer",** Tata McGraw Hill Education, 10th Edition, 2017.
4. F. White, **"Heat and Mass Transfer",** Pearson Education (US), 1988.
5. S. P. Sukhatme, **"A Textbook of Heat Transfer",** Universities Press, 4th Edition, 2005.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME302T					Dynamics of Machines					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the effect of static and dynamic forces on the behaviour of a body/system.
2. To comprehend the function, motions and force distribution in elements like gears, brakes etc.
3. To develop an ability to analyse the motions in various types of gear trains.
4. To determine the unbalance in a system and subsequently balance it.

UNIT I: STATIC AND DYNAMIC FORCE ANALYSIS 10 Hrs.

Static Force Analysis: Static equilibrium, equilibrium for two force and three force members, torque, force convention, free body diagrams, superposition, virtual work. **Dynamic Force Analysis:** D 'Alembert's principle, dynamic analysis of four link, slider crank, velocity and acceleration of automobile components, turning moment, dynamically equivalent system, turning moment diagrams, fluctuation of energy, flywheels.

UNIT II: GEAR AND GEAR TRAINS 10 Hrs.

Gears: Laws of gearing, gears terminology, tooth form, standard interchangeable tooth profile, minimum number of teeth on pinion in contact with a gear, interference and undercutting, bevel, helical and spiral gears. **Gear Trains:** Simple, compound, reverted and epicyclic gear trains, analytical and tabular methods, torques in epicyclic train

UNIT III: BALANCING 10 Hrs.

Balancing: Balancing of rotating masses in single and multiple planes. **Balancing of Reciprocating Masses:** Primary and secondary balancing of reciprocating masses. Unbalanced forces and couples examination of "V", multi cylinder inline engines for primary and secondary balancing.

UNIT IV: VIBRATION 10 Hrs.

Brakes and Dynamometers: Introduction, general description and study of operating principle. **Vibration**: Types of vibrations, free longitudinal and transverse vibrations, whirling speeds, damped vibrations, logarithmic decrement, forced vibrations, free torsional vibrations of single, two and three rotor system.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall force analysis basics, gear terminology, and principles.
CO2 : Understand how force analysis, gear systems including gear trains within the context of kinematics.
CO3 : Apply force analysis to solve engineering problems and design gear systems.
CO4 : Analyze mechanical systems to solve vibration issues, assessing balancing techniques.
CO5 : Evaluate the effectiveness of balancing methods, gear train designs, and related components.
CO6 : Analyze various types of vibrations to optimize machinery performance.

TEXT/REFERENCE BOOKS

1. S. S. Ratan "**Theory of Machines"**, Tata Mc Graw-Hill publications, New Delhi.
2. Ghosh Amitabha, "**Theory of Mechanisms and Machines"**, East West Press.
3. Rao J.S. and Dukkipati R.V, "**Mechanisms and Machines Theory"**, Wiley Eastern Ltd.
4. Shigley J.E and Uicker J.J, **"Theory of Mechanisms and Machines**", Oxford University Press.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME302P					Dynamics of Machines Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To analyze the behavior of bodies undergoing vibrations, both theoretically as well as practically.
2. To statically and dynamically balance an unbalanced rotating system.
3. To calculate gyroscopic torques in a gyroscope apparatus.
4. To compare the behavior of various configurations of governors.

LIST OF EXPERIMENTS

1. To study Undamped free vibrations (longitudinal) of a spring mass system
2. To study Undamped free vibrations (torsional) of a single rotor system
3. To determine the natural frequency of a two rotor system
4. To determine the Gyroscopic couple and its effect on a rotating disc
5. To demonstrate the effect of static and dynamic unbalance in a system.
6. To completely balance a system of rotating masses.
7. To find the damping coefficient of a system undergoing torsional oscillations
8. To plot the characteristics of Forced Damped Vibrations for the given system
9. To demonstrate whirling of shafts
10. To determine the characteristics of governors
11. To determine the jump speed of a Cam-follower system
12. To determine the radius of gyration of a body using Bi-filar and Tri-filar Suspension.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Analyse** the behaviour of bodies undergoing vibrations, both theoretically as well as practically
CO2 : **Apply** the principles of force and couple balance to statically and dynamically balance an unbalanced rotating system
CO3 : **Compute** the damping coefficient of system subjected to transverse and torsional vibrations.
CO4 : **Compare and assess** the dynamic behaviour of various mechanisms
CO5 : **Understand** the principles of gyroscope and governors
CO6 : **Illustrate and apply** a method to compute moment of inertia of rigid bodies

TEXT/REFERENCE BOOKS

1. Dynamics of Machines Lab Manual.
2. SS Rattan, **"Theory of machines",** Tata McGraw-Hill Education, New Delhi.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME303T					Manufacturing Process-I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To provide the basics and advances in manufacturing processes.
2. To provide the fundamentals and working principles of casting, welding, and metal working processes
3. To provide the basics of defects, causes and remedies for different manufacturing processes.
4. To provide overview of novel manufacturing techniques for processing materials such as ceramics, plastics.

UNIT I: CASTING 10 Hrs.

Introduction to manufacturing-classification-challenges-selection. Introduction to casting, foundries, types of castings-patterns-allowances-moulding sand and ingredients-gating system-cores-chaplets-crucibles. Making of-Patterns-Cores-Casting design. Solidification of casting. Casting of metals-plastics. Casting Types-Sand Casting-Permanent-Investment-Die-Centrifugal-Continuous-CO2 Casting-Evaporative pattern. Casting defects, prediction and remedies,

UNIT II: METAL WORKING PROCESSES 10 Hrs.

Classification and concepts of metal working processes. **Forging**-Principles-Classifications-Tools-Dies-Defects. **Rolling**- Fundamentals-Theory-Tools-Types-Defects.**Extrusion**-Classifications-Tools-Dies-Operations-Defects. **Drawing**-Wires-Bars-Tubes-Forces. **Sheet metal forming-**Bending- Deep drawing-Roll forming- Stamping-Spinning-Peen forming. Defects in forging, causes and remedies.

UNIT III: WELDING 10 Hrs.

Introduction-Welding-Soldering-Brazing. Classification of welding-fluxes. Gas welding-ARC- MIG-TIG-Resistance- Defects-Causes-Remedies. Destructive and NDT of welds.

UNIT IV: OTHER MANUFACTURING PROCESSES 10 Hrs.

Processing of ceramics: Ceramic powder preparation, synthesis of ceramic powders, fabrication of ceramic products from powders via sintering, Processing of Plastics: thermoplastics and thermosets, Processing of Plastics: Extrusion. Injection moulding. Thermoforming. Compression moulding. Transfer moulding. General Behavior of polymer melts. Additive Manufacturing Overview.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Define the fundamentals of manufacturing processes and their classifications.
CO2 : Understand the working principles of various casting processes.
CO3 : Demonstrate the forging processes and its types
CO4 : Classify various types welding processes and their significance
CO5 : Create a map for defects in different manufacturing process along with their remedies and causes.
CO6 : Explore different process for manufacturing of ceramics and plastics.

TEXT/REFERENCE BOOKS

1. Serope Kalpakjian, "**Manufacturing engineering and Technology**" Wesley Publishing Co.
2. Lindberg R.A, "**Processes and Materials of Manufacture**", Prentice Hall of India (P) Ltd.
3. P.N.Rao, "**Manufacturing & Technology: Foundry Forming and Welding**" Tata McGraw Hill Publications.
4. Chee Kai Chua, Kah Fai Leong, Chu Sing Lim, "**Rapid prototyping: Principles and applications**" World scientific publications.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – V

24ME303P					Manufacturing Process-I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To analyse and understand various manufacturing techniques
2. To understand and analyse types of defects in welding, casting and metal forming processes
3. To understand moulding process and its different types

LIST OF EXPERIMENTS

1. To study the safety aspects of welding, casting and metal forming operations
2. To study the Shielded Metal Arc Welding Process
3. To study the Gas welding and Gas cutting processes
4. To study Gas Tungsten Arc welding process
5. To Study Resistance Spot Welding process and its types.
6. To understand the use of various welding gauges
7. To study Soldering and Brazing process
8. To study the Wire-Arc Additive Manufacturing Process
9. To study Compression moulding process for polymers.
10. Demonstration of sand-casting operations & manufacturing of various cast products
11. To study the Friction welding and Friction stir welding process.
12. To study the Gas Metal Arc Welding Process

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand various Manufacturing processes
CO2 : Analyze the effect of process parameters on different quality parameters
CO3 : Examine Quality of the manufactured product
CO4 : Produce sand cast mould and cast a component
CO5 : Evaluate Additive manufacturing process
CO6 : Understand Solid state welding processes

TEXT/REFERENCE BOOKS

1. P. N. Rao, **"Manufacturing & Technology: Foundry Forming and Welding"**, Tata McGraw Hill Publications.
2. Serope Kalpakjian & Steuen. R. Sechmid, **"Manufacturing Technology"**, Pearson Education Asia
3. Taylor H.F Flemings M.C & Wulff J., "**Foundry Engineering"**, Wiley Eastern Limited